

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Ionic Equilibrium - Core Application	8 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Ionic Equilibrium - Core Application (High Yield — 8 PYQs)

Core Principle: Key problem pattern in Kcet Ionic Equilibrium.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 0.15 mole of pyridinium chloride has been added to 500cm³ of 0.2 M pyridine solution (a base). Assuming there is no change..." **[Ans: N/A]**
- **[2024 Shift] Q#2 (MCQ):** "Solubility product of CaC₂O₄ at a given temperature in pure water is $4 \times 10^{-9}(\text{molL}^{-1})^2$. Solubility of CaC₂O₄..." **[Ans: N/A]**
- **[2023 Shift] Q#3 (MCQ):** "A weak acid with pK_a 5.9 and weak base with pK_b 5.8 are mixed in equal proportions. pH of the resulting solution is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 0.15 mole of pyridinium chloride has been added to 500cm³ of 0.2 M pyridine solution (a base). Assuming there is no change in volume upon mixing (Given: $pK_{(b)}$ of pyridine = 8.82), the pH of the resulting solution is

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Correct Answer: None

Solution Analysis: Moles of pyridine present initially: $0.2 \times (500 / 1000) = 0.1 \text{ mol}$ Moles of pyridinium chloride added: 0.15 mol Pyridine is a weak base, and pyridinium chloride gives its conjugate acid, pyridinium ion. So the mixture acts as a basic buffer. For a basic buffer, $pOH = pK_{(b)} + \log([salt] / [base])$ Since volume is the same for both, we can directly use moles: $pOH = 8.82 + \log(0.15 / 0.10)$ $pOH = 8.82 + \log(1.5)$ $\log(1.5) \approx 0.18$ So, $pOH = 8.82 + 0.18 = 9.00$ Now, $pH = 14 - pOH$ $pH = 14 - 9 = 5$ So the pH of the result